

EXPERIMENT NO. 5

Title: Apply Array Methods and Object Handling; Create a Cart Total Calculator with Discount Logic.

Software/Tools Required

JavaScript

HTML

CSS

Visual Studio Code / Code Editor

Web Browser

Theory

Arrays, Objects, and Cart Total Calculation in JavaScript

JavaScript provides powerful data structures such as arrays and objects for storing and managing data efficiently. Arrays allow multiple values to be stored in a single variable, while objects store data in the form of key-value pairs. JavaScript also provides various built-in array methods that simplify data manipulation and processing.

1. Arrays in JavaScript An array is a collection of multiple values stored under a single variable name. Each element is identified using an index starting from 0.

2. Array Methods

- a) **push()** Adds an element to the end of an array.
- b) **pop()** Removes the last element from an array.
- c) **map()** Creates a new array by applying a function to every element.
- d) **filter()** Creates a new array containing elements that satisfy a condition.
- e) **reduce()** Reduces an array to a single value.
- f) **shift()** Removes the first element from an array.
- g) **unshift()** Adds an element to the beginning of an array.
- h) **splice()** Adds or removes elements from a specific position.
- i) **slice()** Returns a portion of an array without modifying the original array.
- j) **forEach()** Executes a function for each element of an array.

3. Objects in JavaScript An object is a collection of properties represented as key-value pairs. Objects are useful for storing related information about real-world entities.

Case Study 1

Case Study Title: JavaScript Array Methods Demonstration

Problem Statement

Create an interactive webpage to demonstrate different JavaScript array methods such as push, pop, shift, unshift, splice, slice, map, filter, reduce, forEach, max, and min.

Implementation

The program creates an array containing numbers and provides buttons to perform different array operations. Each button calls a JavaScript function that performs the selected operation and displays the updated array or result dynamically on the webpage. The program also calculates the maximum and minimum values using Math.max() and Math.min().

Conclusion

Arrays and objects are fundamental data structures in JavaScript that enable efficient storage and manipulation of data. Array methods such as map(), filter(), and reduce() simplify data processing tasks, while objects help organize related information. By combining these concepts, JavaScript can be used effectively in practical applications.